Regularized Multivariate Two-way Functional Principal Component Analysis

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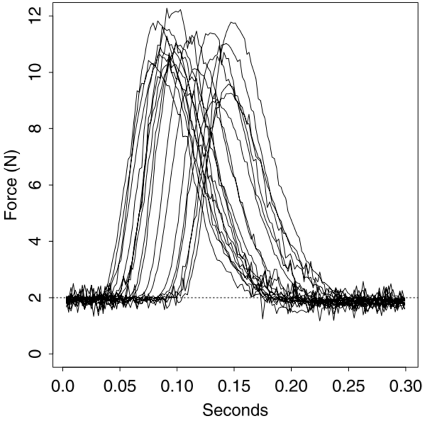
Outline

1. Introduction & Background
2. A SVD Approach for Regularized Multivariate FPCA
3. Two-way Regularized Multivariate FPCA
4. Conclusion & Future Work

Introduction

Functional Data Analysis:

- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.



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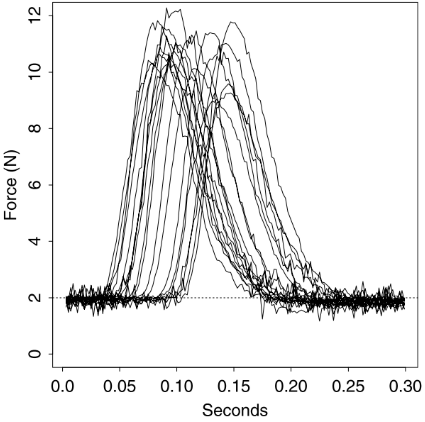

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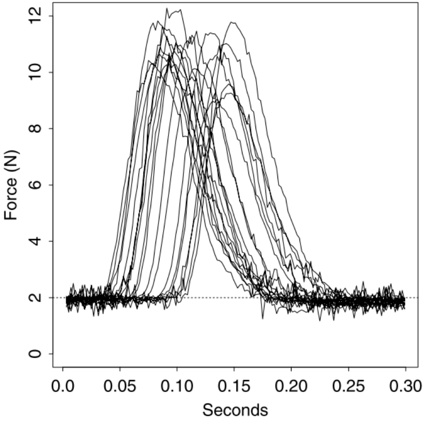

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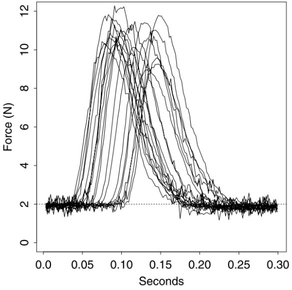

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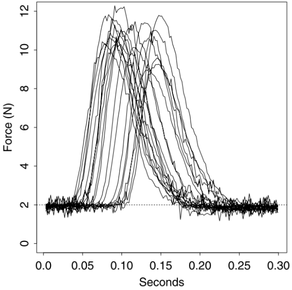

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Regularized MFPCA

Two-way Regularized MFPCA

Conclusion & Future Work